STAT 20000: Elementary Statistics

Homework 5

Due: 11:59 pm, May 15, 2026

All page, section, and exercise numbers below refer to the course text *Statistics*, 4th edition, by D. Freedman, R. Pisani & R. Purves (FPP).

Reading: Chapters 16–20 in FPP.

Q1 (6 points). (Chapter 16, Review Exercise 4, on p.285)

(a) A die will be rolled some number of times, and you win \$1 if it shows an ace (•) more than 20% of the time. Which is better: 60 rolls, or 600 rolls? Explain.

(b) As in (a), but you win the dollar if the percentage of aces is more than 15%.

(c) As in (a), but you win the dollar if the percentage of aces is between 15% and 20%.

Q2 (5 points). (Chapter 17, Review Exercise 7, on p.305) One hundred draws are made at random with replacement from the box $\boxed{1\ \ 2\ \ 3\ \ 4\ \ 5\ \ 6}$.

(a) If the sum of the draws is 321, what is their average?

(b) If the average of the draws is 3.78, what is the sum?

(c) Estimate the chance that the average of the draws is between 3 and 4.

Q3 (4 points). (Chapter 17, Review Exercise 8, on p.305) A coin is tossed 100 times.

(a) The difference "number of heads − number of tails" is like the sum of 100 draws from one of the following boxes. Which one, and why?

   (i) $\boxed{\text{heads}\ \ \text{tails}}$

   (ii) $\boxed{-1\ \ 1}$

   (iii) $\boxed{-1\ \ 0}$

   (iv) $\boxed{0\ \ 1}$

   (v) $\boxed{-1\ \ 0\ \ 1}$

(b) Find the expected value and standard error for the difference.

Q4 (4 points). (Chapter 19, Review Exercise 12, on p.353) The *San Francisco Chronicle* reported on a survey of top high-school students in the US. According to the survey:

> *Cheating is pervasive. Nearly 80 percent admitted some dishonesty, such as copying someone's homework or cheating on an exam.* The survey was sent last spring to 5,000 of the nearly 700,000 high achievers included in the 1993 edition of *Who's Who Among American High School Students*. The results were based on the 1,957 completed surveys that were returned. "The survey does not pretend to be representative of all teenagers," said *Who's Who* spokesman Andrew Weinstein. "Students are listed in *Who's Who* if they are nominated by their teachers or guidance counselors. Ninety-eight percent of them go on to college."

(a) Why isn't the survey "representative of all teenagers"?

(b) Is the survey representative "of the nearly 700,000 high achievers included in the 1993 edition of *Who's Who Among American High School Students*"? Answer yes or no, and explain briefly.

Q5 (2 points). (Chapter 16, Review Exercise 9, on p.286) A box contains red and blue marbles; there are more red marbles than blue ones. Marbles are drawn one at a time from the box, at random with replacement. You win a dollar if a red marble is drawn more often than a blue one. There are two choices:

(A) 100 draws are made from the box.

(B) 200 draws are made from the box.

Choose one of the four options below and explain your answer.

(i) A gives a better chance of winning.

(ii) B gives a better chance of winning.

(iii) A and B give the same chance of winning.

(iv) Can't tell without more information.

Q6 (2 points). (Chapter 18, Review Exercise 7, on p.328) A pair of dice are thrown. Is the total number of spots like

(i) one draw from the box $\boxed{2\ \ 3\ \ 4\ \ 5\ \ 6\ \ 7\ \ 8\ \ 9\ \ 10\ \ 11\ \ 12}$, or

(ii) the sum of two draws from the box $\boxed{1\ \ 2\ \ 3\ \ 4\ \ 5\ \ 6}$?

Explain which option is correct.

Q7 (2 points). (Chapter 19, Review Exercise 5, on p.351) (Hypothetical) A survey is carried out by the finance department to determine the distribution of household size in a certain city. They draw a simple random sample of 1,000 households. After several visits, the interviewers find people at home in only 653 of the sample households. Rather than face such a high non-response rate, the department draws a second batch of households, and uses the first 347 completed interviews in the second batch to bring the sample up to its planned strength of 1,000 households. The department counts 3,087 people in these 1,000 households, and estimates the average household size in the city to be about 3.1 persons. Is this estimate likely to be too low, too high, or about right? Why?